

Seeding Markov Chain Monte Carlo algorithms

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Abstract

The purpose of this work is to illustrate the need of seeding Markov Chain Monte Carlo algorithms when dealing with unobserved variables, as well as to propose a seeding procedure based on a distance-based approach for such instances.

1 Introduction

In real-world scenarios, we often aim to explain a set of traits of a phenomenon of interest under specific conditions. From a probabilistic perspective, we usually propose a stochastic mechanism to approximate the underlying law that is governing the phenomenon of interest in such a way that the outcomes of such a stochastic mechanism can be regarded as realisations of the set of traits, say $\pi(\mathbf{z}|\boldsymbol{\theta})$, where $\mathbf{z}$ is the set of traits of interest and $\boldsymbol{\theta}$ is a random vector that fully defines the proposed stochastic mechanism.

The aforementioned proposal is the core of Statistics, a probabilistic framework for drawing statements about the set of traits by leveraging available measurements of them, say $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N$, and denoted by $\mathbf{z}_{1:N}$. In this context, there are two approaches that prevail. On the one hand, the frequentist approach focuses on estimating $\boldsymbol{\theta}$, say $\hat{\boldsymbol{\theta}}$. On the other hand, the Bayesian approach (Bernardo and Smith, 1994) requires the elicitation of $\pi(\boldsymbol{\theta})$, also known as the prior distribution, but returns via Bayes' rule $\pi(\boldsymbol{\theta}|\mathbf{z}_{1:N})$, also known as the posterior distribution.

We provide a brief overview of the current state-of-the-art MCMC software in Section 2. We then address the problem of seeding MCMC algorithms when dealing with unobserved variables and describe the proposed seeding algorithm for overcoming such scenarios in Section 3, whereas in Section 4 we illustrate it with a case study of Ecology using the R packages **NIMBLE** (de Valpine et al., 2026) and **cmdstanr** (Gabry et al., 2025). Finally, in Section 5 we conclude with....

2 MCMC in practice

Markov Chain Monte Carlo (MCMC) methods (Brooks et al., 2011) are widely used in practice for reconstructing posterior distributions. These methods rely on the theoretical principle of defining a homogeneous, Harris recurrent, and aperiodic Markov chain whose stationary probability density function is the target posterior distribution, that is, to construct a suitable Markov chain $\tilde{\boldsymbol{\theta}}^{(1)}, \tilde{\boldsymbol{\theta}}^{(2)}, \dots$ such that $\tilde{\boldsymbol{\theta}}^{(t)} \underset{t \rightarrow \infty}{\sim} \pi(\boldsymbol{\theta} | \mathbf{z}_{1:N})$. See Meyn and Tweedie (2009) for a concise treatment of the theory of Markov chains.

Historically, the BUGS project (Lunn et al., 2012) led the efforts to enable Bayesian methods and JAGS (Plummer, 2003) as one BUGS engine.

Lately, Hamiltonian Monte Carlo (HMC) methods (Neal, 2011) have become an alternative to MCMC methods.

HMC is the workhorse of the Stan software (Stan Development Team, 2026). On the other hand, the NIMBLE system (de Valpine et al., 2017).

3 Seeding MCMC algorithms

A more challenging scenario is when we only observed a subset of $\mathbf{z}$, that is, if $\mathbf{z} = (\mathbf{x}, \mathbf{y})$ we only have measurements for $\mathbf{y}$, say $\mathbf{y}^{obs}$. In this case, we instead seek to reconstruct $\pi(\mathbf{x}, \boldsymbol{\theta} | \mathbf{y}^{obs})$, and thus, we now also have to initialise $\mathbf{x}$. Built upon a distance-based approach (Maa et al., 1996) in tandem with the prior predictive distribution, we proposed the following seeding process for Monte Carlo algorithms when dealing with unobserved variables.

$d : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a pseudo-distance. Algorithm 1 summaries such a seeding process.

Algorithm 1 Distance-based Seeding

Require: $\theta^{(t)}$ **Ensure:** $\theta^{(t)}, \mathbf{x}^{(t)}, \mathbf{y}^{(t)}$

- 1: $t = 0$
 - 2: **repeat**
 - 3: $t \leftarrow t + 1$
 - 4: $\theta^{(t)} \sim \pi(\theta)$
 - 5: $\mathbf{x}^{(t)}, \mathbf{y}^{(t)} \sim \pi(\mathbf{x}, \mathbf{y} | \theta)$
 - 6: **until** $d(\mathbf{y}^{(t)}, \mathbf{y}^{obs}) < d^{max}$
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4 Case study: Population with exponential growth

[Auger-Méthé et al. \(2021\)](#) provides an overview of Bayesian methods in Ecology, whereas [Monnahan et al. \(2017\)](#) propose to use HMC for making inference.

For a discrete-time setting, let us assume that a population experience an exponential growth with time-varying rate r_t , that is, $N_{t+1} = e^{r_t} N_t$ for $t = 1, 2, \dots, T - 1$. We can rewrite the time-varying rate as $r_t = f_t - m_t$, where $f_t, m_t \in \mathbb{R}^+$ are the fertility and mortality rates at time t , respectively. Now, at each time t , we can think of abundance before and after mortality, say N_t^{pre} and N_t^{post} , respectively. Thus, if M_t denotes the total mortality we have that $N_t^{post} = N_t^{pre} - M_t$.

Given the aforementioned population demographics, we can propose the following stochastic population mechanism:

$$\begin{aligned} M_t &\sim \text{Bin}(N_t^{pre}, 1 - e^{-m_t}) \\ N_{t+1}^{pre} &\sim \text{Poi}(e^{f_t} N_t^{post}) \end{aligned}$$

Furthermore,

$$\begin{aligned} \log(N_1^{pre}) &\sim \text{Poi}(e^{LN_0}) \\ \log(a_t) &\sim \text{N}(\mu_{la}, \sigma_{la}) \\ \log(b_t) &\sim \text{N}(\mu_{lb}, \sigma_{lb}) \\ \log(f_t) &\sim \text{N}(\mu_{lf}, \sigma_{lf}) \end{aligned}$$

with

$$\begin{aligned}\mu_{lm} &\sim N(m_{lm}, sd_{lm}) \\ \sigma_{lm} &\sim N(0, sd_{lm}) \\ \mu_{lf} &\sim N(m_{lf}, sd_{lf}) \\ \sigma_{lf} &\sim N(0, sd_{lf})\end{aligned}$$

The R project ([R Core Team, 2026](#)).

5 Concluding Remarks

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